Good alternatives to the standard tools

The standard tools are the ones you will meet most often. But for many steps there is another tool that is **faster, simpler, or better for a different situation**. This page lists the ones worth knowing about, in the order of a data science project.

> **How to read it:** each row says which standard tool the new one replaces or sits next to. "Better" depends on the problem: choose by your data size, your team and your deadline.

---

1. Processing data

Replaces / next to	Try	Why
pandas	**Polars**	Much faster and lighter on memory, with *lazy* queries that are planned before they run
pandas	**DuckDB**	Write SQL directly on CSV, Parquet or a DataFrame, with no server
looking at tables in a notebook	**DBeaver**	A free app to browse and query data with SQL
pandas	**PySpark**	The same ideas on a cluster, for data too big for one machine

2. Versioning and environments

Replaces / next to	Try	Why
GitHub	**GitLab, Codeberg**	Other hosts for the same git repository
Git (for big files)	**DVC**	Git for **data and models**: git keeps a small pointer, the big file lives in storage
GitHub + MLflow	**DagsHub**	Code, data and experiments in one place
pip + venv	**uv**	Very fast, creates the environment and locks exact versions

3. Language models

Replaces / next to	Try	Why
`transformers` (running a model)	**Ollama, LM Studio**	Run a model on your own computer with one command
`transformers` (many requests)	**vLLM**	A server that answers many requests quickly
bitsandbytes (quantization)	**llama.cpp (GGUF), AWQ, GPTQ**	Other ways to shrink a model, from laptop CPUs to GPUs

Replaces / next to	Try	Why
parsing the model's text by hand	**Pydantic, Instructor**	Force **valid, structured** output, such as a label as JSON
Hugging Face (finding models)	**Ollama library, Kaggle, OpenRouter**	Other places to get or call models

4. Classical machine learning

Replaces / next to	Try	Why
scikit-learn (random forest)	**XGBoost, LightGBM**	Gradient boosting, usually stronger on table data
scikit-learn (big data)	**Spark MLlib**	The same kind of tools on a cluster
scikit-learn (grid search)	**Optuna**	Smarter search for hyperparameters

5. Tracking experiments

Replaces / next to	Try	Why
MLflow	**Weights & Biases**	Polished dashboards, popular in research
MLflow	**TensorBoard**	Simple and free, built into PyTorch and TensorFlow
MLflow	**DVC experiments**	Tracks experiments inside git, with no server

6. Tests, deployment and automation

Replaces / next to	Try	Why
pytest (tests of the code)	**pandera, Great Expectations**	Tests for the **data**: columns, types and ranges
running checks by hand	**Ruff, pre-commit**	Check and format the code before every commit
FastAPI (a quick demo)	**Streamlit, Gradio**	A web demo of your model in a few lines
GitHub Actions (a pipeline)	**Prefect, Airflow**	Run the steps of a pipeline in order, on a schedule

7. Notebooks and annotation

Replaces / next to	Try	Why
Jupyter	**Marimo**	A notebook that is a plain Python file and updates what depends on a change
a spreadsheet for annotation	**Label Studio**	A tool made for annotation, with several annotators

One rule to remember: a newer tool is not automatically a better one. Choose by the size of the problem, by what your team already knows, and by how well the tool is maintained.